

Note 7: QEC3

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7.1 Summary of Progress

1. Generalize operational properties for Pauli operators, projection operators and Hadamard transform.
2. Explored theoretical construction, error correction mechanism and circuit decomposition of CSS codes.
3. Studied simplicial homology, $\mathbb{F}_4$ representation and topological origin of quantum codes.

$$\left\{ \begin{array}{c} \text{surface} \\ \text{code} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{topological} \\ \text{code} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{CSS} \\ \text{code} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{stabilizer} \\ \text{code} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{Quantum} \\ \text{code} \end{array} \right\}$$

7.2 Further study of CSS code

Theorem 1 Let k be a field with $\text{char } k = 0$ and

1. $\mathcal{H}$ be a vector space over k ;
2. G be a finite group of $\text{GL}(\mathcal{H})$;
3. The subspace

$$\mathcal{H}^G := \{v \in \mathcal{H} \mid \forall g \in G, gv = v\};$$

then the operator

$$P_G := \frac{1}{|G|} \sum_{g \in G} g : \mathcal{H} \rightarrow \mathcal{H}$$

is a projector with $\mathcal{H}^G = \text{Im } P_G$.

Lemma 2 Let C be a linear code, then

$$\frac{1}{|C|} \sum_{y \in C} (-1)^{x \cdot y} = \chi_{C^\perp}(x)$$

Lemma 3 For $a, b \in \mathbb{F}_2^n$, denote $X^a = X^{a_1} \otimes \dots \otimes X^{a_n}$, $Z^b := Z^{b_1} \otimes \dots \otimes Z^{b_n}$, then we have

$$X^a |x\rangle = |x + a\rangle, \quad Z^b |y\rangle = (-1)^{b \cdot y} |y\rangle.$$

Lemma 4 Let H be the Hadamard matrix and $H_n := H^{\otimes n}$, $a, b, w \in \mathbb{F}_2^n$, then

- $H^2 = I$, $H^\dagger = H = H^{-1}$;
- $HX = ZH$, $HZ = XH$;
- $H_n^2 = I$, $H_n^\dagger = H_n = H_n^{-1}$;
- $H_n X^a = Z^a H_n$, $H_n Z^b = X^b H_n$.

Proof Recall the single-qubit Hadamard matrix and Pauli matrices:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

First, calculate H^2 :

$$H^2 = \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \right) \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \right) = \frac{1}{2} \begin{pmatrix} 1+1 & 1-1 \\ 1-1 & 1+1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

Since H is real symmetric, $H^\dagger = H$. Combined with $H^2 = I$, we have $H^{-1} = H$.

Next, verify $HX = ZH$:

$$\begin{aligned} HX &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0+1 & 1+0 \\ 0-1 & 1+0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}. \\ ZH &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1+0 & 1+0 \\ 0-1 & 0+1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}. \end{aligned}$$

For $H_n = H^{\otimes n}$ (n-qubit Hadamard), we have:

$$H_n^2 = (H^{\otimes n})^2 = H^2 \otimes H^2 \otimes \cdots \otimes H^2 = I \otimes I \otimes \cdots \otimes I = I.$$

Since H_n is real symmetric, $H_n^\dagger = H_n$, so $H_n^{-1} = H_n$.

Let $a = (a_1, a_2, \dots, a_n) \in \mathbb{F}_2^n$, where $a_i \in \{0, 1\}$. From the single-qubit case $HX^{a_i} = Z^{a_i}H$, we have:

$$\begin{aligned} H_n X^a &= (H \otimes H \otimes \cdots \otimes H) \cdot (X^{a_1} \otimes X^{a_2} \otimes \cdots \otimes X^{a_n}) \\ &= (HX^{a_1}) \otimes (HX^{a_2}) \otimes \cdots \otimes (HX^{a_n}) \\ &= (Z^{a_1}H) \otimes (Z^{a_2}H) \otimes \cdots \otimes (Z^{a_n}H) \\ &= (Z^{a_1} \otimes Z^{a_2} \otimes \cdots \otimes Z^{a_n}) \cdot (H \otimes H \otimes \cdots \otimes H) \\ &= Z^a H_n. \end{aligned}$$

Similarly, $H_n Z^b = X^b H_n$ for any $b \in \mathbb{F}_2^n$. ■

Corollary 5 Let $|+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{\epsilon \in \mathbb{F}_2^n} |\epsilon\rangle$ be the uniform superposition state. For any $w \in \mathbb{F}_2^n$,

$$H^{\otimes n} |w\rangle = Z^w |+\rangle^{\otimes n}.$$

Proof By the Hadamard lemma,

$$H^{\otimes n} |w\rangle = \frac{1}{\sqrt{2^n}} \sum_{\epsilon \in \mathbb{F}_2^n} (-1)^{w \cdot \epsilon} |\epsilon\rangle.$$

By the PauliZ lemma, $Z^w |\epsilon\rangle = (-1)^{w \cdot \epsilon} |\epsilon\rangle$, so

$$H^{\otimes n} |w\rangle = \frac{1}{\sqrt{2^n}} \sum_{\epsilon} Z^w |\epsilon\rangle = Z^w \left(\frac{1}{\sqrt{2^n}} \sum_{\epsilon} |\epsilon\rangle \right) = Z^w |+\rangle^{\otimes n}.$$
■

Error (noise) for CSS code

We encode $|x + C_2\rangle$ by

$$|\psi\rangle = |x + C_2\rangle = \frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} |x + y\rangle$$

Affected by noise $\mathcal{E} = X^{e_1} Z^{e_2}$ with $e_1, e_2 \in \mathbb{F}_2^n$, the state is changed to

$$|\psi_0\rangle = \mathcal{E}|\psi\rangle = \frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} (-1)^{(x+y) \cdot e_2} |x + y + e_1\rangle.$$

Recover operation for CSS code

The recover process can be expressed by the chain

$$\begin{aligned} |\psi_0\rangle &\mapsto |\psi_0\rangle \otimes |\vec{0}\rangle \xrightarrow{U_{H_1}} |\psi_1\rangle \xrightarrow{\text{tr}_B} |\psi_2\rangle \xrightarrow{X^{e_1}} |\psi_3\rangle \xrightarrow{H^{\otimes n}} |\psi_4\rangle \\ &\mapsto |\psi_4\rangle \otimes |\vec{0}\rangle \xrightarrow{U_{H_2}} |\psi_5\rangle \xrightarrow{\text{tr}_B} |\psi_6\rangle \xrightarrow{X^{e_2}} |\psi_7\rangle \xrightarrow{H^{\otimes n}} |\psi\rangle \end{aligned}$$

Where U_{H_1}, U_{H_2} is given by the lemma: Let $\mathcal{H}_A = (\mathbb{C}^2)^{\otimes n}$ and $\mathcal{H}_B = (\mathbb{C}^2)^{\otimes r}$.

Lemma 6 Given $H \in \mathbb{F}_2^{r \times n}$, there is a unitary operation U_H such that

$$U_H(|c\rangle \otimes |0^{\otimes r}\rangle) = |c\rangle \otimes |Hc\rangle, \quad \forall c \in \mathbb{F}_2^n.$$

Proof The total Hilbert space is $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$, whose standard orthonormal basis is

$$\{|c\rangle \otimes |d\rangle \mid c \in \mathbb{F}_2^n, d \in \mathbb{F}_2^r\}.$$

We define the linear operator U_H on the standard basis of the whole Hilbert space as

$$U_H(|c\rangle \otimes |d\rangle) = |c\rangle \otimes |d + Hc\rangle,$$

It suffices to show this map is a permutation of all basis states. Suppose

$$|c_1\rangle \otimes |d_1 + Hc_1\rangle = |c_2\rangle \otimes |d_2 + Hc_2\rangle.$$

Tensor product basis vectors are linearly independent, so we have $c_1 = c_2$ and $d_1 + Hc_1 = d_2 + Hc_2$.

Since $c_1 = c_2$, we deduce $d_1 = d_2$. Hence the mapping is injective. The collection of all basis vectors is a finite set, so injectivity implies surjectivity.

Therefore U_H is bijective on the full basis set, which means it is a permutation of orthonormal bases. ■

Remark 7 We adopt the PVM measurement $\{\Pi_s \mid s \in \mathbb{F}_2^r\}$ acting on the ancilla subsystem, where

$$\Pi_s := I_A \otimes |s\rangle\langle s|.$$

This measurement enables extraction of error information. Furthermore, taking partial trace over ancilla subsystem yields

$$\text{tr}_B(U_H(\cdot \otimes |0^{\otimes r}\rangle)) = I_A.$$

Theorem 8 Let error operator $\mathcal{E} = X^{e_1} Z^{e_2}$ and define recovery operation

$$\mathcal{R} = H^{\otimes n} X^{e_2} \text{tr}_B (U_{H_2}(\cdot \otimes |0^{\otimes r}\rangle)) H^{\otimes n} X^{e_1} \text{tr}_B (U_{H_1}(\cdot \otimes |0^{\otimes r}\rangle)),$$

then the composition satisfies $\mathcal{R} \circ \mathcal{E}(|\psi\rangle) = |\psi\rangle$.

Proof

$$\begin{aligned} \mathcal{R} \circ \mathcal{E}(|\psi\rangle) &= H^{\otimes n} X^{e_2} \text{tr}_B (U_{H_2}(\cdot \otimes |0^{\otimes r}\rangle)) H^{\otimes n} X^{e_1} \text{tr}_B (U_{H_1}(\cdot \otimes |0^{\otimes r}\rangle)) (X^{e_1} Z^{e_2} |\psi\rangle) \\ &= H^{\otimes n} X^{e_2} \text{tr}_B (U_{H_2}(\cdot \otimes |0^{\otimes r}\rangle)) H^{\otimes n} X^{e_1} X^{e_1} Z^{e_2} |\psi\rangle \\ &= H^{\otimes n} X^{e_2} \text{tr}_B (U_{H_2}(\cdot \otimes |0^{\otimes r}\rangle)) H^{\otimes n} Z^{e_2} |\psi\rangle \\ &= H^{\otimes n} X^{e_2} \text{tr}_B (U_{H_2}(\cdot \otimes |0^{\otimes r}\rangle)) X^{e_2} H^{\otimes n} |\psi\rangle \\ &= H^{\otimes n} X^{e_2} X^{e_2} H^{\otimes n} |\psi\rangle \\ &= H^{\otimes n} H^{\otimes n} |\psi\rangle \\ &= |\psi\rangle \end{aligned}$$

■

Quantum Circuit for recover operation

How to decompose U_{H_1} and U_{H_2} into a product of elementary quantum gates? We have the following lemma.

Lemma 9 Let $H = (h_{ij}) \in \mathbb{F}_2^{r \times n}$. Let $q_1, \dots, q_n$ be qubits of the system $\mathcal{H}_A$, and $p_1, \dots, p_r$ be qubits of the ancilla system $\mathcal{H}_B$. The unitary operator admits decomposition

$$U_H = \prod_{1 \leq i \leq r, 1 \leq j \leq n, h_{ij}=1} \text{CNOT}_{q_j \rightarrow p_i},$$

where $\text{CNOT}_{q_j \rightarrow p_i}$ uses qubit q_j as control and qubit p_i as target.

Let $H = \sum_{i=1}^r \sum_{j=1}^n h_{ij} E_{ij}$, then

$$\begin{aligned} U_{H(i,j)} (|c\rangle \otimes |d\rangle) &= |c\rangle \otimes |d + h_{ij} E_{ij} c\rangle \\ U_{H(1,2)} U_{H(1,1)} (|c\rangle \otimes |d\rangle) &= U_{H(1,2)} (|c\rangle \otimes |d + h_{11} E_{11} c\rangle) \\ &= |c\rangle \otimes |(d + h_{11} E_{11} c) + h_{12} E_{12} c\rangle \end{aligned}$$

by induction

$$\begin{aligned} \prod_{i=1}^r \prod_{j=1}^n U_{H(i,j)} (|c\rangle \otimes |d\rangle) &= |c\rangle \otimes \left| d + \sum_{i=1}^r \sum_{j=1}^n h_{ij} E_{ij} c \right\rangle \\ &= |c\rangle \otimes |d + Hc\rangle \end{aligned}$$

So

$$\prod_{i=1}^r \prod_{j=1}^n U_{H(i,j)} = U_H.$$

Moreover,

$$U_{H(i,j)} (|c\rangle \otimes |d\rangle) = \begin{cases} |c\rangle \otimes |d + E_{ij} c\rangle & h_{ij} = 1 \\ |c\rangle \otimes |d\rangle & h_{ij} = 0 \end{cases}.$$

If $h_{ij} = 1$, then

$$U_{H(i,j)}(|c_1 c_2 \dots c_n\rangle \otimes |d_1 \dots d_r\rangle) = |c_1 \dots c_n\rangle \otimes |d_1 \dots (d_i + c_j) \dots d_r\rangle.$$

Thus $U_{H(i,j)} = \text{CNOT}_{q_j \rightarrow p_i}$, is a CNOT gate.

Example 10 Take parameters $n = 3$, $r = 2$, binary matrix

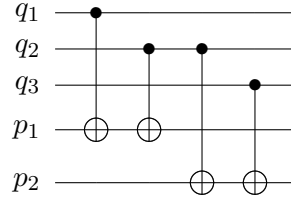
$$H = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \in \mathbb{F}_2^{2 \times 3}$$

Let principal qubits be q_1, q_2, q_3 , ancilla qubits be p_1, p_2 . The binary vector is $\mathbf{c} = (c_1, c_2, c_3) \in \mathbb{F}_2^3$.

All entries satisfying $h_{ij} = 1$ are $(1, 1), (1, 2), (2, 2), (2, 3)$. The unitary decomposition reads:

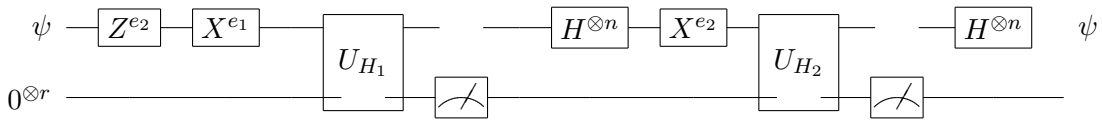
$$U_H = \text{CNOT}_{q_1 \rightarrow p_1} \cdot \text{CNOT}_{q_2 \rightarrow p_1} \cdot \text{CNOT}_{q_2 \rightarrow p_2} \cdot \text{CNOT}_{q_3 \rightarrow p_2}$$

Quantum circuit diagram:



For general case :

$$\mathcal{R} \circ \mathcal{E} = H^{\otimes n} X^{e_2} \text{Tr}_B \left(\prod_{h_{ij}^{(2)}=1} \text{CNOT}_{q_j \rightarrow p_i} \right) H^{\otimes n} X^{e_1} \text{Tr}_B \left(\prod_{h_{ij}^{(1)}=1} \text{CNOT}_{q_j \rightarrow p_i} \right) X^{e_1} Z^{e_2}$$



7.3 Topological code

Definition 11 Let $V_1, V_2 \subseteq \mathbb{F}_2^n$ be linear subspaces satisfying $V_1 \perp V_2$. The CSS code is the stabilizer code with stabilizer group

$$\mathcal{G} = \{X^a Z^b \mid (a, b) \in V_1 \times V_2 \subseteq \mathbb{F}_2^{2n}\}.$$

Lemma 12 If $H_X \in \mathbb{F}_2^{m_X \times n}$, $H_Z \in \mathbb{F}_2^{m_Z \times n}$ with $m_X, m_Z \leq n$, then

$$H_X H_Z^T = 0 \iff C_X \perp C_Z,$$

where $C_X = \text{Im} H_X^T$, $C_Z = \text{Im} H_Z^T$.

Proof : $H_X H_Z^T = 0 \iff C_Z = \text{Im} H_Z^T \subseteq \text{Ker} H_X$. It suffices to prove $\text{Ker} H_X \perp \text{Im} H_X^T = C_X$.

- Consider the linear map

$$A : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n = \mathbb{F}_2^{m_X} \oplus \mathbb{F}_2^{n-m_X}, \quad x \mapsto \begin{pmatrix} H_X x \\ 0 \end{pmatrix} = \begin{pmatrix} H_X \\ 0 \end{pmatrix} x;$$

then

$$A = \begin{pmatrix} H_X \\ 0 \end{pmatrix}, \quad A^T = (H_X^T \ 0).$$

- The adjoint map is

$$A^* : \mathbb{F}_2^n = \mathbb{F}_2^{m_X} \oplus \mathbb{F}_2^{n-m_X} \rightarrow \mathbb{F}_2^n, \quad \begin{pmatrix} x \\ y \end{pmatrix} \mapsto (H_X^T \ 0) \begin{pmatrix} x \\ y \end{pmatrix} = H_X^T x$$

- Using the lemma, we have

$$\begin{aligned} \text{Im} H_X^T &= \text{Im} A^* = (\text{Ker} A)^\perp \\ &= \left\{ x \in \mathbb{F}_2^n \mid \begin{pmatrix} H_X x \\ 0 \end{pmatrix} = 0 \right\}^\perp \\ &= (\text{Ker} H_X)^\perp = C_X^\perp \end{aligned}$$

The proof is complete. ■

$$\begin{aligned} \{X^a | a \in V_1\} &\leftrightarrow V_1 = C_X = \text{Im} H_X^T \\ \{Z^b | a \in V_2\} &\leftrightarrow V_2 = C_Z = \text{Im} H_Z^T \\ V_1 \perp V_2 &\leftrightarrow H_X H_Z^T = 0 \leftrightarrow H_Z H_X^T = 0 \end{aligned}$$

Remark 13 In other word,

$$x + C_2 \in C_1/C_2 = \text{Ker} H_X / \text{Im} H_Z^T = H_1(K)$$

belong to homological group of K , where K denote the chain complex

$$\mathbb{F}_2^{m_Z} \xrightarrow{H_Z^T} \mathbb{F}_2^n \xrightarrow{H_X} \mathbb{F}_2^{m_X}.$$

In general, if you give me a chain complex

$$C_2(K; \mathbb{F}_2) \xrightarrow{\partial_2} C_1(K; \mathbb{F}_2) \xrightarrow{\partial_1} C_0(K; \mathbb{F}_2),$$

i.e. $\partial_1 \circ \partial_2 = 0$. then

$$H_X = \partial_1, H_Z = \partial_2^T$$

define a CSS code.

Corollary 14 If K is a simplicial complexes 7.5.3, then $\partial_{i-1} \circ \partial_i = 0$ for all $i \geq 0$, thus

$$V_1 = \text{Im} \partial_{i-1}^T, V_2 = \text{Im} \partial_i$$

define a CSS code.

7.4 Plan for next week

1. Textbook[1] chap.10 and chap. 4.
2. study surface code and its physical realization .
3. study symplectic self-orthogonal code and Grassmanian varieties.

7.5 Appendix

7.5.1 GF(4) check for symplectic product

Lemma 15 We have the following isomorphisms as field GF(4)

$$\mathbb{F}_4 \simeq \frac{\mathbb{F}_2[x]}{\langle x^2 + x + 1 \rangle} \simeq \mathbb{F}_2[\omega] = \{a + b\omega : a, b \in \mathbb{F}_2\}$$

while $\omega \neq 0, 1$ and $\omega^2 = \omega + 1$. So the field is unique up to isomorphism.

Proof Since $\mathbb{F}_4/\mathbb{F}_2$ is a field extension with $[\mathbb{F}_4 : \mathbb{F}_2] = 2$, there is $\omega \in \mathbb{F}_4$ such that

$$\mathbb{F}_4 = \mathbb{F}_2[\omega] = \{a + b\omega : a, b \in \mathbb{F}_2\}.$$

Then

$$\mathbb{F}_4 = \mathbb{F}_2[\omega] = \{0, 1, \omega, \omega^2\} = \{0, 1, \omega, \omega + 1\},$$

Thus $\omega^2 = \omega + 1$. Consider the ring homomorphism

$$\phi : \mathbb{F}_2[x] \rightarrow \mathbb{F}_4, \quad f(x) \mapsto f(\omega),$$

We have $\mathbb{F}_4 = \text{im } \phi \simeq \mathbb{F}_2[x]/\ker \phi$. And for $f \in \ker \phi$. There is $q(x), r(x) \in \mathbb{F}_2[x]$

$$f(x) = (x^2 + x + 1)q(x) + r(x), \quad \deg r(x) < 2.$$

$$0 = f(\omega) = (\omega^2 + \omega + 1)q(\omega) + r(\omega) = r(\omega)$$

So we have $r(x) = 0 \implies \ker \phi = \langle x^2 + x + 1 \rangle$, a maximal ideal of $\mathbb{F}_2[x]$. ■

Corollary 16 $\omega \mapsto \omega^2 = \omega + 1$ induces a field automorphism of $\mathbb{F}_4$.

Definition 17 For $u = a + b\omega, a, b \in \mathbb{F}_2$, define

1. Conjugate element

$$\bar{u} := a + b\omega^2 = a + b + b\omega.$$

2. Trace

$$\text{tr}(u) := \text{Tr}_{\mathbb{F}_2}(u) := u + \bar{u} = b$$

Remark 18 $u + \bar{u} = a + b\omega + a + b\omega^2 = b(\omega + \omega^2) = b$.

Proposition 19 If $(a, b), (c, d) \in \mathbb{F}_2^2, u = a + b\omega, v = c + d\omega$, then

$$\text{tr}(u\bar{v}) = \text{tr}(\bar{u}v) = ad + bc.$$

Proof Proved by formula $\text{tr}(a_0 + b_0\omega) = b_0$ and

$$\begin{aligned} u\bar{v} &= (a + b\omega)(c + d\omega^2) \\ &= ac + ad\omega^2 + bc\omega + bd \\ &= ac + bd + ad(\omega + 1) + bc\omega \\ &= ac + bd + ad + (ad + bc)\omega \end{aligned}$$
■

Corollary 20 *Let the correspondence be*

$$I \leftrightarrow 0, \quad X \leftrightarrow 1, \quad Z \leftrightarrow \omega, \quad Y \leftrightarrow \omega^2.$$

For Pauli operators

$$P = P_1 \otimes P_2 \otimes \cdots \otimes P_n, \quad Q = Q_1 \otimes Q_2 \otimes \cdots \otimes Q_n,$$

let $P_j \leftrightarrow u_j$ and $Q_j \leftrightarrow v_j$ in $\mathbb{F}_4$. Then

$$[P, Q] = 0 \iff \sum_{j=1}^n \text{tr}(u_j \bar{v}_j) = 0.$$

Proof We have known that if $P = X^a Z^b, Q = X^c Z^d$, then

$$[P, Q] = 0 \iff a \cdot d + b \cdot c = 0 \iff \sum_{j=1}^n a_j d_j + b_j c_j = 0.$$

which is equivalent to $u_j = a_j + b_j \omega, v_j = c_j + d_j \omega$ and

$$\sum_{j=1}^n \text{tr}(u_j \bar{v}_j) = 0.$$

This completes the proof. ■

7.5.2 Some supplementary of materials for linear algebra

Lemma 21 *Consider finite dimensional vector space V over a field k . Given a **non-degenerate** bilinear form $B : V \times V \rightarrow k$ a non-, then for any linear map $A : V \rightarrow V$, we have a unique linear map $A^* : V \rightarrow V$ such that*

$$B(Av, w) = B(v, A^*w) : \forall v, w \in V.$$

Remark 22 . We say A^* is the adjoint map of A .

Theorem 23 *For any linear map $A : V \rightarrow V$, we have*

$$(\text{Im} A)^\perp = \text{Ker} A^*, \quad (\text{Ker} A)^\perp = \text{Im} A^*;$$

$$V = \text{Ker} A^* \oplus^\perp \text{Im} A = \text{Ker} A \oplus^\perp \text{Im} A^*;$$

where $W^\perp := \{v \in V | \forall w \in W, B(v, w) = 0\}$.

Proof For $B : V \times V \rightarrow k$ is non-degenerate, we have $B(V, w) = \{0\} \implies w = 0$. So

$$\begin{aligned} (\text{Im} A)^\perp &= \{v \in V | \forall w \in V, B(Aw, v) = 0\} \\ &= \{v \in V | \forall w \in V, B(w, A^*v) = 0\} \\ &= \{v \in V | A^*v = 0\} \\ &= \text{Ker} A^*. \end{aligned}$$

Similarly we can prove that $(\text{Ker} A)^\perp = \text{Im} A^*$. ■

Corollary 24 If $A^* = A$ or $A^* = -A$, then

$$V = \text{Ker} A \oplus^\perp \text{Im} A.$$

Theorem 25 The following statements are equivalent:

1. $V = \text{Ker} A \oplus \text{Im} A$;
2. $V = \text{Ker} A + \text{Im} A$;
3. $\text{Ker} A \cap \text{Im} A = \{0\}$;
4. $\text{Ker} A = \text{Ker} A^2$;
5. $\text{Im} A = \text{Im} A^2$.
6. $\text{Im} A = W, A|_W : W \rightarrow W$ is invertible.

Proof

- $1 \Rightarrow 2$ is obvious.
- $2 \Rightarrow 3$: By dimension formulas

$$\dim V = \dim \text{Ker} A + \dim \text{Im} A$$

and

$$\dim (\text{Ker} A + \text{Im} A) = \dim \text{Ker} A + \dim \text{Im} A - \dim (\text{Ker} A \cap \text{Im} A),$$

we have $\dim (\text{Ker} A \cap \text{Im} A) = 0$.

- $3 \Rightarrow 4$: $\text{Ker} A \subseteq \text{Ker} A^2$ is obvious.
If $A^2 x = 0$, then $Ax \in \text{Ker} A \cap \text{Im} A = \{0\}$. So $Ax = 0 \implies x \in \text{Ker} A$.
- $4 \Rightarrow 5$: $\text{Im} A^2 \subseteq \text{Im} A$ is obvious. For the dimension formulas

$$\begin{aligned} \dim V &= \dim \text{Im} A + \dim \text{Ker} A, \\ \dim V &= \dim \text{Im} A^2 + \dim \text{Ker} A^2, \end{aligned}$$

we have $\dim \text{Im} A^2 = \dim \text{Im} A$.

- Since $\text{Im} A = \text{Im} A^2 = W$, so $A|_W : W \rightarrow W$ is surjective, so is injective for $\dim V < \infty$.
- $6 \Rightarrow 5$: For $A|_W : W \rightarrow W$ is invertible, $\dim AW = \dim W$. So

$$\dim \text{Im} A^2 = \dim A^2 V = \dim AW = \dim W = \dim \text{Im} A.$$

- $5 \Rightarrow 4$: Similar to $4 \Rightarrow 5$.
- $5, 4 \Rightarrow 3$: If $x \in \text{Ker} A \cap \text{Im} A$, then

$$Ax = 0, x = Ay \implies A^2 y = 0 \implies y \in \text{Ker} A^2 = \text{Ker} A \implies x = Ay = 0.$$

- $3 \Rightarrow 1$: By formula

$$\dim (\text{Ker} A + \text{Im} A) = \dim \text{Ker} A + \dim \text{Im} A - \dim (\text{Ker} A \cap \text{Im} A) = \dim V - 0 = \dim V.$$

■

Corollary 26 If $A^* = A$ or $A^* = -A$, then A satisfies all the equivalent conditions above.

7.5.3 Simplicial Complexes and Simplicial Homology Groups over $\mathbb{F}_2$

Definition 27 (Simplex and Face) An n -simplex $\underline{s} = (a_0, a_1, \dots, a_n)$ is the convex hull of $n + 1$ affinely independent points $a_0, \dots, a_n$ in Euclidean space, called *vertices*. A simplex $\underline{t}$ is a **face** of $\underline{s}$ ($\underline{t} < \underline{s}$) if all vertices of $\underline{t}$ are vertices of $\underline{s}$.

Definition 28 (Boundary of a Simplex) The **boundary** $\partial \underline{s}$ of an n -simplex $\underline{s}$ is the union of all its proper faces. Geometrically, $\partial \underline{s} \cong S^{n-1}$.

Definition 29 (Oriented Simplex) Let $\underline{s} = (a_0, a_1, \dots, a_n)$ be an n -simplex. An *oriented n -simplex* s is defined as

$$s = [a_0, a_1, \dots, a_n], \quad -s = [a_{\sigma(0)}, a_{\sigma(1)}, \dots, a_{\sigma(n)}]$$

where σ is an **odd permutation** of indices. Two orderings define the same orientation if related by an even permutation.

Definition 30 (Simplicial Complex) A *simplicial complex* K is a set of simplices such that:

1. Any two simplices are either disjoint or intersect in a common face.
2. If $s \in K$, then all faces of s are also in K .

Definition 31 ($\mathbb{F}_2$ -Chain Group) The q -dimensional chain group $C_q(K; \mathbb{F}_2)$ is the $\mathbb{F}_2$ -vector space generated by all oriented q -simplices of K . We write $N_q = \dim_{\mathbb{F}_2} C_q(K; \mathbb{F}_2)$.

Remark 32 For $-1 = 1$ over $\mathbb{F}_2$, $-s = s$, so **orientation is irrelevant**.

Example 33 Define the simplicial complex

$$K = \{[a_0], [a_1], [a_2], [a_0, a_1], [a_0, a_2], [a_1, a_2], [a_0, a_1, a_2]\}.$$

1. The vector space $C_2(K; \mathbb{F}_2)$ is generated by the unique 2-simplex $[a_0, a_1, a_2]$, so its basis is $\{[a_0, a_1, a_2]\}$ and $N_2 = \dim_{\mathbb{F}_2} C_2(K; \mathbb{F}_2) = 1$.
2. The vector space $C_1(K; \mathbb{F}_2)$ is generated by the three 1-simplices $[a_1, a_2], [a_0, a_2], [a_0, a_1]$, so its basis is $\{[a_1, a_2], [a_0, a_2], [a_0, a_1]\}$ and $N_1 = \dim_{\mathbb{F}_2} C_1(K; \mathbb{F}_2) = 3$.

Definition 34 (Boundary Homomorphism over $\mathbb{F}_2$) The boundary homomorphism

$$\partial_q : C_q(K; \mathbb{F}_2) \rightarrow C_{q-1}(K; \mathbb{F}_2)$$

is defined on an oriented q -simplex $s = [a_0, a_1, \dots, a_q]$ by

$$\partial_q s := \sum_{i=0}^q (-1)^i [a_0, \dots, \hat{a}_i, \dots, a_q] = \sum_{i=0}^q [a_0, \dots, \hat{a}_i, \dots, a_q]$$

and extended linearly over $\mathbb{F}_2$.

Remark 35 Let $N_q = \dim_{\mathbb{F}_2} C_q(K; \mathbb{F}_2)$, $N_{q-1} = \dim_{\mathbb{F}_2} C_{q-1}(K; \mathbb{F}_2)$. For a general q -chain

$$c = \sum_{j=1}^{N_q} \lambda_j s_j \in C_q(K; \mathbb{F}_2)$$

with coefficients $\lambda_j \in \mathbb{F}_2$, the linear extension is given by

$$\partial_q c = \sum_{j=1}^{N_q} \lambda_j \partial_q s_j.$$

And we can get a chain of morphism

$$\dots \xrightarrow{\partial_{q+1}} C_q(K) \xrightarrow{\partial_q} C_{q-1}(K) \xrightarrow{\partial_{q-1}} \dots \xrightarrow{\partial_2} C_1(K) \xrightarrow{\partial_1} C_0(K).$$

Example 36 Define the simplicial complex

$$K = \{ [a_0], [a_1], [a_2], [a_0, a_1], [a_0, a_2], [a_1, a_2], [a_0, a_1, a_2] \}.$$

By the definition of boundary homomorphism over $\mathbb{F}_2$,

$$\partial_2[a_0, a_1, a_2] = \sum_{i=0}^2 [a_0, \dots, \hat{a}_i, \dots, a_2] = [a_1, a_2] + [a_0, a_2] + [a_0, a_1].$$

According to the matrix representation of boundary homomorphism, the boundary matrix $\partial_2 \in \mathbb{F}_2^{3 \times 1}$ is

$$\partial_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Represent the 2-chain $[a_0, a_1, a_2]$ as the vector $\mathbf{c} = (1) \in \mathbb{F}_2^1$, then

$$\partial_2[a_0, a_1, a_2] = \partial_2 \mathbf{c} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Matrix Representation of Boundary Homomorphism is given by the proposition: Let $N_q = \dim_{\mathbb{F}_2} C_q(K; \mathbb{F}_2)$

Proposition 37 Suppose $C_q(K; \mathbb{F}_2)$ has basis $\{s_1, \dots, s_{N_q}\}$ and $C_{q-1}(K; \mathbb{F}_2)$ has basis $\{t_1, \dots, t_{N_{q-1}}\}$. The homomorphism ∂_q corresponds to a boundary matrix $M \in \mathbb{F}_2^{N_{q-1} \times N_q}$ with entries

$$M_{ij} = \begin{cases} 1 & t_i \text{ is contained in } \partial_q s_j, \\ 0 & \text{otherwise.} \end{cases}$$

For any chain $c \in C_q(K; \mathbb{F}_2)$ represented by a vector $\mathbf{c} \in \mathbb{F}_2^{N_q}$, we have

$$\partial_q c = M \mathbf{c}.$$

Proof Since $\{s_1, \dots, s_{N_q}\}$ is a basis of $C_q(K; \mathbb{F}_2)$, any q -chain can be uniquely written as

$$c = \sum_{j=1}^{N_q} c_j s_j, \quad \mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_{N_q} \end{pmatrix} \in \mathbb{F}_2^{N_q}.$$

By linearity of ∂_q over $\mathbb{F}_2$,

$$\partial_q c = \sum_{j=1}^{N_q} c_j \partial_q s_j.$$

For each basis element $s_j \in C_q$, expand $\partial_q s_j$ in the basis $\{t_1, \dots, t_{N_{q-1}}\}$ of C_{q-1} :

$$\partial_q s_j = \sum_{i=1}^{N_{q-1}} M_{ij} t_i,$$

where $M_{ij} = 1$ if $t_i \in \partial_q s_j$, else $M_{ij} = 0$, by definition of the matrix entries. Substitute into the expression for $\partial_q c$:

$$\partial_q c = \sum_{j=1}^{N_q} c_j \left(\sum_{i=1}^{N_{q-1}} M_{ij} t_i \right) = \sum_{i=1}^{N_{q-1}} \left(\sum_{j=1}^{N_q} M_{ij} c_j \right) t_i.$$

The coefficient vector of this linear combination is exactly $M\mathbf{c} \in \mathbb{F}_2^{N_{q-1}}$. Therefore, $\partial_q c = M\mathbf{c}$. ■

Definition 38 ($\mathbb{F}_2$ -Homology Group) The q -dimensional homology group over $\mathbb{F}_2$ is

$$H_q(K; \mathbb{F}_2) = \ker \partial_q / \text{Im } \partial_{q+1}.$$

Theorem 39 (Homotopy Invariance of Homology Groups) Let K, L be finite simplicial complexes. If their geometric realizations $|K|$ and $|L|$ are homotopy equivalent, then for every integer $q \geq 0$, the $\mathbb{F}_2$ -homology groups are isomorphic:

$$H_q(K; \mathbb{F}_2) \cong H_q(L; \mathbb{F}_2).$$

Definition 40 (Betti Number over $\mathbb{F}_2$) The q -th Betti number over $\mathbb{F}_2$ is $\beta_q = \dim_{\mathbb{F}_2} H_q(K; \mathbb{F}_2)$.

Definition 41 (Euler Characteristic) Let K be a finite simplicial complex, $N_q = \dim_{\mathbb{F}_2} C_q(K; \mathbb{F}_2)$ the number of q -simplices in K . The *Euler characteristic* of K is defined as

$$\chi(K) = \sum_{q=0}^{\dim K} (-1)^q N_q.$$

Theorem 42 (Euler-Poincaré Formula & Homotopy Invariance) Let K be a finite simplicial complex, $\beta_q = \dim_{\mathbb{F}_2} H_q(K; \mathbb{F}_2)$ the q -th $\mathbb{F}_2$ -Betti number. Then

$$\chi(K) = \sum_{q=0}^{\dim K} (-1)^q \beta_q.$$

Moreover, Euler characteristic is a **homotopy invariant**.

Proof Let $n = \dim K$. By ranknullity theorem over $\mathbb{F}_2$,

$$N_q = \dim C_q = \dim \ker \partial_q + \dim \text{Im } \partial_q.$$

From the short exact sequence $0 \rightarrow \text{Im } \partial_{q+1} \rightarrow \ker \partial_q \rightarrow H_q \rightarrow 0$,

$$\dim \ker \partial_q = \dim \text{Im } \partial_{q+1} + \beta_q.$$

Substitute into the Euler characteristic:

$$\begin{aligned} \chi(K) &= \sum_{q=0}^n (-1)^q N_q = \sum_{q=0}^n (-1)^q (\dim \text{Im } \partial_{q+1} + \beta_q + \dim \text{Im } \partial_q) \\ &= \sum_{q=0}^n (-1)^q \beta_q + \sum_{q=0}^n (-1)^q \dim \text{Im } \partial_{q+1} + \sum_{q=0}^n (-1)^q \dim \text{Im } \partial_q. \end{aligned}$$

Reindexing the first image sum shows the two boundary image terms cancel exactly, hence

$$\chi(K) = \sum_{q=0}^n (-1)^q \beta_q.$$

■

7.5.4 PVM (Projective-Valued Measurement)

Postulate 1 (Observable and Measurement) Every observable corresponds to a self-adjoint operator $A = A^\dagger$. For discrete spectra:

$$A = \sum_k \lambda_k |\lambda_k\rangle\langle\lambda_k|,$$

where $\lambda_k \in \mathbb{R}$ are eigenvalues (observed value). For a state $|\psi\rangle$, the probability to measure λ_k is

$$p(\lambda_k) = |\langle\lambda_k|\psi\rangle|^2.$$

After measurement, the state collapses to $|\lambda_k\rangle$.

Example 43 (Eigenstate Deterministic Measurement) Take observable $A = |0\rangle\langle 0|$, its spectral decomposition is

$$A = 1 \cdot |0\rangle\langle 0| + 0 \cdot |1\rangle\langle 1|.$$

For normalized state $|\psi\rangle = a|0\rangle + b|1\rangle$ with $|a|^2 + |b|^2 = 1$. By Born rule:

$$p(1) = |\langle 0|\psi\rangle|^2 = |a|^2, \quad p(0) = |\langle 1|\psi\rangle|^2 = |b|^2.$$

All measurement outcomes satisfy probability normalization. Only when $|\psi\rangle = |0\rangle$, we obtain deterministic result $p(1) = 1$. Tiny deviation component will lead to uncertain measurement results.

Definition 44 (PVM) A **projective-valued measurement (PVM)** on a Hilbert space $\mathcal{H}$ is a set of orthogonal projection operators $\{\Pi_i\}$ satisfying:

1. $\Pi_i^2 = \Pi_i$, $\Pi_i^\dagger = \Pi_i$;
2. $\Pi_i \Pi_j = 0$ for $i \neq j$;
3. $\sum_i \Pi_i = I$.

Remark 45 The projector Π_k can be a high-dimensional subspace projection, such as $\Pi = |0\rangle\langle 0| + |1\rangle\langle 1|$, which corresponds to degenerate eigenspace with identical eigenvalue.

Proposition 46 (Pure State Probability for PVM) Let $\{\Pi_i\}$ be a PVM on $\mathcal{H}$, and let $|\psi\rangle \in \mathcal{H}$ be a pure state. The probability of obtaining outcome corresponding to Π_k is

$$p_k := p(\text{outcome} \in \text{Im } \Pi_k) = \langle\psi|\Pi_k|\psi\rangle.$$

Proof Projection operator satisfies $\Pi_k^2 = \Pi_k$, $\Pi_k^\dagger = \Pi_k$, whose eigenvalues are merely 0 and 1.

Suppose the eigenspace with eigenvalue 1 admits an orthonormal basis $\{|e_j\rangle\}_j$.

By the measurement postulate, the total probability that the state collapses into this subspace equals the sum of probabilities over all basis components:

$$p_k = \sum_j |\langle e_j|\psi\rangle|^2.$$

Expand the squared modulus and rearrange the summation:

$$\begin{aligned} p_k &= \sum_j \langle\psi|e_j\rangle\langle e_j|\psi\rangle \\ &= \left\langle\psi\left|\left(\sum_j |e_j\rangle\langle e_j|\right)\right|\psi\right\rangle. \end{aligned}$$

Note that the projector onto this subspace is exactly

$$\Pi_k = \sum_j |e_j\rangle\langle e_j|,$$

substitute to obtain

$$p_k = \langle \psi | \Pi_k | \psi \rangle.$$

This completes the proof. ■

Proposition 47 (Mixed State Probability for PVM) *Let $\{\Pi_i\}$ be a PVM on $\mathcal{H}$, and let ρ be a density operator describing mixed state. The probability of measurement outcome falling into $\text{Im } \Pi_k$ satisfies*

$$p_k = p(\text{outcome} \in \text{Im } \Pi_k) = \text{tr}(\rho \Pi_k).$$

Proof Any mixed state density operator can be decomposed into convex combination of pure states

$$\rho = \sum_m c_m |\psi_m\rangle\langle\psi_m|, \quad c_m \geq 0, \quad \sum_m c_m = 1.$$

From the proven result of pure state measurement probability:

$$p_k^{(m)} = \langle \psi_m | \Pi_k | \psi_m \rangle.$$

Probability obeys linear superposition for mixed ensemble:

$$p_k = \sum_m c_m \cdot \langle \psi_m | \Pi_k | \psi_m \rangle.$$

Utilize cyclic property and linearity of trace:

$$\begin{aligned} \langle \psi_m | \Pi_k | \psi_m \rangle &= \text{tr}(|\psi_m\rangle\langle\psi_m| \Pi_k), \\ p_k &= \sum_m c_m \text{tr}(|\psi_m\rangle\langle\psi_m| \Pi_k) \\ &= \text{tr}\left(\left(\sum_m c_m |\psi_m\rangle\langle\psi_m|\right) \Pi_k\right) \\ &= \text{tr}(\rho \Pi_k). \end{aligned}$$

This completes the proof for mixed state. ■

Remark 48 Consider bipartite Hilbert space $\mathcal{H}_S \otimes \mathcal{H}_A$, where $\mathcal{H}_S$ denotes the principal system and $\mathcal{H}_A$ the ancilla subsystem. Let U_H be the global unitary operator acting on the composite system. The mapping $U_H(\cdot \otimes |0\rangle_A)$ characterizes unitary evolution after embedding the principal state into the joint system with initialized ancilla.

Carry out complete PVM measurement $\{\Pi_k^A\}$ on $\mathcal{H}_A$. Summing over all mutually orthogonal projective outcomes and performing partial trace tr_A over ancilla degrees of freedom, the statistical averaging procedure cancels the interactive disturbance imposed on $\mathcal{H}_S$.

The resultant reduced effective operator coincides with identity operator on $\mathcal{H}_S$, satisfying

$$\text{tr}_A(U_H(\cdot \otimes |0\rangle_A)) = I_S.$$

References

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